pushing with his finger on the world tube of a watch's second hand. His pushes produce a backward-in-time gravitational force, which makes the second-hand twitch in a Morse-encoded pattern that carries the quantum data. The tesseract stores the twitching pattern in the bulk so it repeats over and over again. When forty-year-old Murph returns to her bedroom three decades later, she finds the second hand still twitching, repeating over and over again the encoded quantum data that Cooper has struggled so hard to send her.

How does the backward-in-time gravitational force work? I'll describe my physicist's interpretation after I tell you what I know, or think I know, about backward time travel.

Time Travel Without a Bulk: What I Think I Know



In 1987, triggered by Carl Sagan (Chapter 14), I realized something amazing about wormholes. If wormholes are allowed by the laws of physics, then Einstein's relativistic laws permit transforming them into time machines. The nicest example of this was discovered a year later by my close friend Igor Novikov, in Moscow, Russia. Igor's example, Figure 30.3, shows that a wormhole's transformation into a time machine might occur naturally, without the aid of intelligent beings.

In Figure 30.3, the bottom mouth of the wormhole is in orbit around a black hole and the upper mouth is far from the black hole. Because of the black hole's intense gravitational pull, Einstein's law of time warps dictates that time flow more slowly at the lower mouth than at the upper mouth. More slowly, that is, when compared along the path of gravity's intense pull: the dashed purple path through the external universe. I presume, for concreteness, that this has produced a one-hour lag so when compared through the external universe, the bottom clock shown